TEXORA · LECTURE SHEET

Circular Rib Machine

Add a horizontal dial of needles above the cylinder, gate the two sets alternately, and time them so they never collide: the rib machine knits balanced, highly elastic double fabric — and, by needle selection, the whole rib family from 1×1 cuffs to cardigan and Milano structures.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 5 · LEVEL 3 TERM 1

1. Construction: cylinder, dial and gating

The rib machine carries the familiar vertical **cylinder** and, above it, a horizontal disc — the **dial** — with radial tricks holding latch needles that point outward and slide toward and away from the centre. Dial needles knit the reverse loops; cylinder needles knit the face loops. The beds are **gated** (phased) for rib: each dial needle sits *between* two cylinder needles (half-pitch offset), so the two sets interleave without touching as both draw yarn from the same feeder. Dial height above the cylinder, dial timing (whether dial needles draw slightly before, with, or after cylinder needles — **delayed, synchronised or advanced timing**) and the two stitch cams (one per bed) are the machine's master adjustments. Delayed timing (cylinder knits first) is usual for cotton rib: it eases yarn demand at knock-over and reduces press-offs.

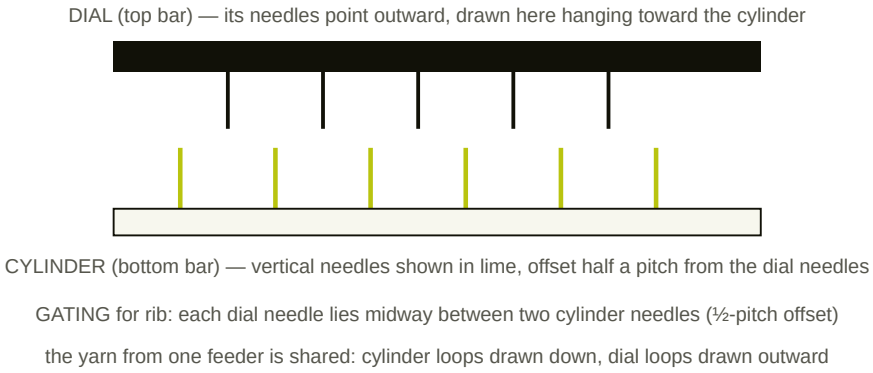


Figure 1 — Cylinder-and-dial arrangement, schematic radial section. Rib gating = half-pitch offset between the beds; interlock (Topic 6) instead gates dial needles directly behind cylinder needles.

2. Knitting action at a feeder

Both beds clear together as their butts ride their clearing cams. The feeder lays yarn across cylinder hooks and dial hooks in one pass. The stitch cams then withdraw the needles — cylinder needles

downward, dial needles inward — each drawing its loop through the previous loop of its own bed, while the yarn segment between a cylinder loop and the next dial loop becomes the connecting sinker loop of the rib course. Because one yarn feeds two beds, the stitch lengths of the two beds interact: tightening the dial stitch cam robs yarn from the cylinder loops. Balance is set by adjusting both cams against measured loop lengths in each bed, not by eye. Rib machines normally run fewer feeders and lower rpm than S/J frames of the same diameter (two cam tracks, more needles in action): a typical 30-inch E18 rib frame carries ≈ 48–60 feeders at 16–22 rpm.

3. Structures by selection

STRUCTURE	NEEDLE/CAM ARRANGEMENT	CHARACTER
1×1 rib	All needles knit every feeder	Maximum widthwise recovery; cuffs, collars
2×2 rib	Pairs of needles active, pairs idle in each bed (needles removed or selected out)	Bolder ribs, sweater bands
Half cardigan	Tuck on dial at one feeder, knit all at next	Fuller, wider than 1×1
Full cardigan	Tuck alternately on each bed	Fullest rib, least lengthwise stretch
Milano rib	Course sequence: rib course, cylinder-only course, dial-only course	Stable, woven-like double fabric for structured garments
Swiss/French piqué (on rib base)	Tuck/miss selection per pattern	Textured polo and dress fabrics

4. Production — worked example (verified)

WORKED EXAMPLE — 1×1 RIB PRODUCTION

Machine: D = 30 in, E18, F = 48 feeders, 20 rpm, stitch length 5.5 mm in each bed, yarn 29.5 tex, efficiency 80 %.

Needles per bed N = π × 30 × 18 = **1696**. One rib course uses both beds: yarn per course = 2 × 1696 × 5.5 mm = 18.66 m.

Courses/min = 48 × 20 = 960. Yarn rate = 960 × 18.66 = 17,914 m/min = 1,074,816 m/h.

Mass = 1074.8 km × 29.5 g/km = 31.71 kg/h theoretical; × 0.80 = **25.4 kg/h actual**.

Compare Topic 4: the rib frame makes fewer, heavier courses — similar kg/h to the jersey machine but far fewer m/h, because each course carries two beds of loops.

5. Faults specific to rib

- **Broken or bent dial needles** → missing reverse wales, holes that show only when the fabric is stretched.
- **Timing slip** → needle collision (cylinder vs dial), bursts of broken needles; always re-time after dial removal.
- **Uneven bed balance** → one face dominates, fabric twists or shows grin-through; correct via the two stitch cams with measured loop lengths.
- **Dial height error** → loops drawn too tight or slack at knock-over, press-offs or excessive yarn demand.

Glossary

Dial — horizontal needle disc above the cylinder. **Gating** — the angular phase between dial and cylinder needles. **Timing** — relative knock-over order of the two beds. **Grin-through** — the reverse loops showing between face wales when rib is stretched. **Milano rib** — rib plus one course on each bed alone per repeat.

Key points

• Rib = cylinder + dial, half-pitch gating, one yarn shared by two beds per course.
• Two stitch cams must be balanced by measured loop length per bed; delayed timing suits cotton.
• Verified 30 in/E18/48F/20 rpm output: 25.4 kg/h at 80 % — heavy courses, modest m/h.
• Selection on the same frame yields 2×2, cardigans and Milano.

Practice

1. Recompute the worked example for F = 60 feeders at 18 rpm (same l, tex, efficiency): kg/h.
2. Why must dial needles be withdrawn (knit) slightly differently from cylinder needles when the dial stitch cam is tightened mid-run?
3. State the gating for rib and contrast it with interlock gating (Topic 6 preview).
4. A rib fabric shows the dial loops grinning through at rest. Give two causes and their corrections.
5. Choose structures for (a) a cuff that must recover to half its stretched width, (b) a stable double fabric for a structured dress. Justify each.

Answer key

1. Courses/min = 60 × 18 = 1080; yarn = 1080 × 18.66 = 20,153 m/min = 1,209,168 m/h; mass = 1209.2 × 29.5 = 35.67 kg/h; × 0.80 = **28.5 kg/h**.
2. (Accept reasoned equivalent: tightening the dial cam draws more yarn into dial loops from the shared feed, shortening cylinder loops in the same course; mid-run adjustment therefore unbalances the beds course-by-course and can print as barré until both cams are re-balanced.)
3. Rib gating: dial needle midway between two cylinder needles (half-pitch offset) so wales alternate. Interlock gating: dial needles directly behind cylinder needles (needle-to-needle) so wales coincide and lock.
4. Causes: dial stitch length too long relative to cylinder (dial loops oversized and visible) — re-balance cams by measured lengths; fabric knitted too loose overall (low tightness) — shorten stitch length / raise cover so face wales close over the reverse loops. Also acceptable: wrong gauge-yarn match (yarn too fine for gauge).
5. (a) 1×1 rib — alternating wales give the greatest widthwise extension and recovery. (b) Milano rib — the single-bed courses lock the rib, giving a stable, almost woven hand.

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